



Packet Sniffing (Core)

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Coding Dojo Cybersecurity Boot Camp
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Assignment

- Analyze network packets using Wireshark.
- Identify network scans from a network capture.

Assignment Submission

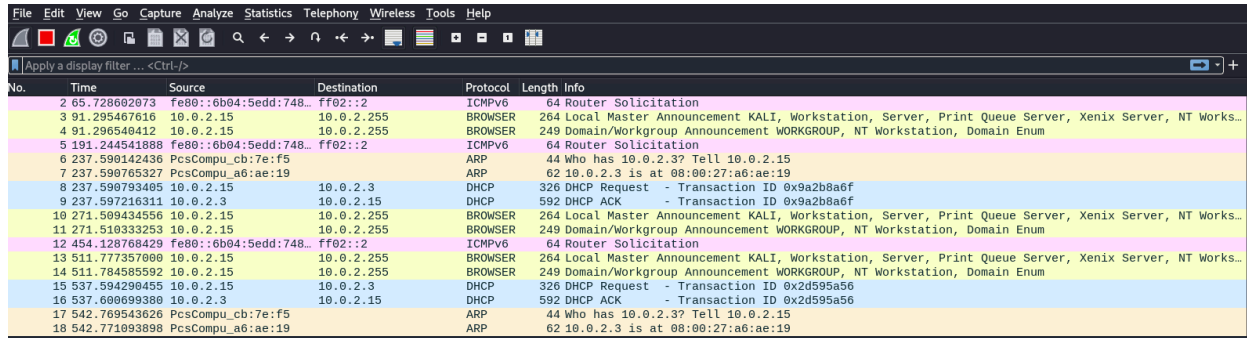
Upload a single document with screenshots of the traffic that Nmap caused. You will not be able to get it all due to the large amount of traffic that you capture during this time. Be sure to provide a sentence or two describing what each screenshot represents.

Assignment Solution:

The **host**, which is the Default gateway of my system is ➔ **192.168.1.254**

The Kali Linux Virtual Server is ➔ **10.0.2.15**

The Wireshark packet sniffing was initiated (*without initiating a network scan, for starters*). Find below the screenshot:



No.	Time	Source	Destination	Protocol	Length	Info
2	65.728662673	fe80::6b04:5edd:748...	ff02::2	ICMPv6	64	Router Solicitation
3	91.295467616	10.0.2.15	10.0.2.255	BROWSER	264	Local Master Announcement KALI, Workstation, Server, Print Queue Server, Xenix Server, NT Works...
4	91.296548412	10.0.2.15	10.0.2.255	BROWSER	249	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum
5	191.244541888	fe80::6b04:5edd:748...	ff02::2	ICMPv6	64	Router Solicitation
6	237.590142436	PcsCompu_cb:7e:f5		ARP	44	Who has 10.0.2.3? Tell 10.0.2.15
7	237.590765327	PcsCompu_a6:ae:19		ARP	62	10.0.2.3 is at 08:00:27:a6:ae:19
8	237.590793405	10.0.2.15	10.0.2.3	DHCP	326	DHCP Request - Transaction ID 0x9a2b8a6f
9	237.597216311	10.0.2.3	10.0.2.15	DHCP	592	DHCP ACK - Transaction ID 0x9a2b8a6f
10	271.569434556	10.0.2.15	10.0.2.255	BROWSER	264	Local Master Announcement KALI, Workstation, Server, Print Queue Server, Xenix Server, NT Works...
11	271.518333253	10.0.2.15	10.0.2.255	BROWSER	249	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum
12	454.328768429	fe80::6b04:5edd:748...	ff02::2	ICMPv6	64	Router Solicitation
13	511.777357900	10.0.2.15	10.0.2.255	BROWSER	264	Local Master Announcement KALI, Workstation, Server, Print Queue Server, Xenix Server, NT Works...
14	511.784585592	10.0.2.15	10.0.2.255	BROWSER	249	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum
15	537.594290455	10.0.2.15	10.0.2.3	DHCP	326	DHCP Request - Transaction ID 0x2d595a56
16	537.600699380	10.0.2.3	10.0.2.15	DHCP	592	DHCP ACK - Transaction ID 0x2d595a56
17	542.769543626	PcsCompu_cb:7e:f5		ARP	44	Who has 10.0.2.3? Tell 10.0.2.15
18	542.771993898	PcsCompu_a6:ae:19		ARP	62	10.0.2.3 is at 08:00:27:a6:ae:19

Observations:

S/No	Wireshark Network Capture Details		Explanation of observation												
1	<table><tr><th>Source</th><th>Destination</th></tr><tr><td>fe80::6b04:5edd:748...</td><td>ff02::2</td></tr></table>		Source	Destination	fe80::6b04:5edd:748...	ff02::2	<table><tr><th>Protocol</th><th>Length</th><th>Info</th></tr><tr><td>ICMPv6</td><td>64</td><td>Router Solicitation</td></tr></table>	Protocol	Length	Info	ICMPv6	64	Router Solicitation		
	Source	Destination													
fe80::6b04:5edd:748...	ff02::2														
Protocol	Length	Info													
ICMPv6	64	Router Solicitation													
	- Source → fe80::6b04:5edd:7483:f856 - Destination → ff02::2 - Protocol → <u>ICMPv6</u> - Type → Router Solicitation <i>The source address is typically the IPv6 address of the host sending the solicitation.</i> <i>The destination address is the all-routers multicast address (ff02::2), indicating that all routers on the local link should receive the message.</i>		The "Router Solicitation" message is sent by an IPv6 host to discover routers on the local link. In IPv6, routers periodically send out "Router Advertisement" messages to announce their presence and provide network configuration information. The "Router Solicitation" is a way for hosts to actively seek out routers.												
S/No	Wireshark Network Capture Details		Explanation of observation												
2	<table><tr><td>10.0.2.15</td><td>10.0.2.255</td><td>BROWSER</td><td>264</td><td>Local Master Announcement KALI, Workstation, Server, Print Queue Server, Xenix Server, NT Works...</td></tr><tr><td>10.0.2.15</td><td>10.0.2.255</td><td>BROWSER</td><td>249</td><td>Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum</td></tr></table>		10.0.2.15	10.0.2.255	BROWSER	264	Local Master Announcement KALI, Workstation, Server, Print Queue Server, Xenix Server, NT Works...	10.0.2.15	10.0.2.255	BROWSER	249	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum			
	10.0.2.15	10.0.2.255	BROWSER	264	Local Master Announcement KALI, Workstation, Server, Print Queue Server, Xenix Server, NT Works...										
10.0.2.15	10.0.2.255	BROWSER	249	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum											
	- Source → 10.0.2.15 - Destination → 10.0.2.255 - Protocol → <u>Browser</u> - Type → Local Master Announcement, etc.		The Kali Linux server 10.0.2.15 is sending a network packet, potentially related to NetBIOS browsing, to all devices on the local network segment (broadcast address 10.0.2.255). This is announcing its presence and sharing information about available resources. (e.g. Print Queue server, etc.)												

S/No	Wireshark Network Capture Details		Explanation of observation	
3	10.0.2.15 10.0.2.3	10.0.2.3 10.0.2.15	DHCP DHCP	326 DHCP Request 592 DHCP ACK
	- Source → 10.0.2.15 - Destination → 10.0.2.3 - Protocol → <u>DHCP</u> - Type → DHCP Request		Dynamic Host configuration protocol request from the source to the destination. There's an acknowledgement from 10.0.2.3 back to 10.0.2.15	
S/N	Wireshark Network Capture Details		Explanation of observation	
4	PcsCompu_cb:7e:f5 PcsCompu_a6:ae:19	ARP ARP	44 Who has 10.0.2.3? Tell 10.0.2.15 62 10.0.2.3 is at 08:00:27:a6:ae:19	
	- Source → PcsCompu_cb:7e:f5 - Destination → - Protocol → <u>ARP</u> - Info → Who has 10.0.2.3? Tell 10.0.2.15		Address Resolution Protocol (ARP) request for an IP address, and indicates for it to be sent to the Kali Linux. Address Resolution Protocol (ARP) reply to the host identifying the indicated IP.	
	- Source → PcsCompu_33:5d:dd - Destination → - Protocol → <u>ARP</u> - Info → 10.0.2.3. is at 08:00:27:33:5d:dd			

While the Wireshark packet sniffing was still ongoing, network scanning using the 'nmap' command was initiated from the Kali Linux VB to my default gateway:
'nmap 192.168.1.254-255'

Find the screenshot below:



No.	Time	Source	Destination	Protocol	Length	Info
41	1749.6949104...	PcsCompu_cb:7e:f5		ARP	44	Who has 10.0.2.3? Tell 10.0.2.15
42	1749.6955988...	PcsCompu_33:5d:dd		ARP	62	10.0.2.3 is at 08:00:27:33:5d:dd
43	1960.4629346...	10.0.2.15	192.168.1.255	ICMP	44	Echo (ping) request id=0xf6ba, seq=0/0, ttl=58 (no response found!)
44	1960.4633055...	10.0.2.15	192.168.1.254	ICMP	44	Echo (ping) request id=0xa71d, seq=0/0, ttl=48 (reply in 57)
45	1960.4635498...	10.0.2.15	192.168.1.255	TCP	60	64829 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
46	1960.4637537...	10.0.2.15	192.168.1.254	TCP	60	64829 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
47	1960.4642784...	10.0.2.15	192.168.1.255	TCP	56	64829 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
48	1960.4646256...	10.0.2.15	192.168.1.254	TCP	56	64829 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
49	1960.4648661...	10.0.2.15	192.168.1.255	ICMP	56	Timestamp request id=0x69e9, seq=0/0, ttl=45
50	1960.4653793...	RealtekU_12:35:00		ARP	62	Who has 10.0.2.15? Tell 10.0.2.1
51	1960.4653910...	PcsCompu_cb:7e:f5		ARP	44	10.0.2.15 is at 08:00:27:cb:7e:f5
52	1960.4653795...	RealtekU_12:35:00		ARP	62	Who has 10.0.2.15? Tell 10.0.2.1
53	1960.4655017...	PcsCompu_cb:7e:f5		ARP	44	10.0.2.15 is at 08:00:27:cb:7e:f5
54	1960.4659121...	10.0.2.15	192.168.1.254	ICMP	56	Timestamp request id=0x6585, seq=0/0, ttl=46
55	1960.4660108...	192.168.1.254	10.0.2.15	TCP	62	80 → 64829 [RST, ACK] Seq=1 Ack=1 Win=32768 Len=0
56	1960.4720065...	192.168.1.254	10.0.2.15	TCP	62	443 → 64829 [SYN, ACK] Seq=0 Ack=1 Win=32768 Len=0 MSS=1460
57	1960.4720067...	192.168.1.254	10.0.2.15	ICMP	62	Echo (ping) reply id=0xa71d, seq=0/0, ttl=63 (request in 44)
58	1960.4720312...	10.0.2.15	192.168.1.254	TCP	56	64829 → 443 [RST] Seq=1 Win=0 Len=0
59	1961.5654466...	10.0.2.15	192.168.1.255	ICMP	56	Timestamp request id=0x915f, seq=0/0, ttl=46
60	1961.5662527...	10.0.2.15	192.168.1.255	TCP	56	64831 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
61	1961.5670944...	192.168.1.255	10.0.2.15	TCP	62	80 → 64831 [RST, ACK] Seq=1 Ack=1 Win=32768 Len=0
62	1961.5672095...	10.0.2.15	192.168.1.255	TCP	60	64831 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
63	1961.5676935...	10.0.2.15	192.168.1.255	ICMP	44	Echo (ping) request id=0x1253, seq=0/0, ttl=46 (no response found!)
64	1961.5990258...	10.0.2.15	192.168.1.254	DNS	88	Standard query 0x9c21 PTR 254.1.168.192.in-addr.arpa
65	1961.5998721...	10.0.2.15	192.168.1.254	DNS	88	Standard query 0x9c21 PTR 255.1.168.192.in-addr.arpa
66	1961.6032321...	192.168.1.254	10.0.2.15	DNS	124	Standard query response 0x9c21 PTR 254.1.168.192.in-addr.arpa PTR dslddevice.attl...
67	1961.6032324...	192.168.1.254	10.0.2.15	DNS	88	Standard query response 0x9c22 No such name PTR 255.1.168.192.in-addr.arpa
68	1961.6234707...	10.0.2.15	192.168.1.255	TCP	60	65085 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
69	1961.6241561...	10.0.2.15	192.168.1.254	TCP	60	65085 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
70	1961.6245278...	10.0.2.15	192.168.1.255	TCP	60	65085 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
71	1961.6256489...	10.0.2.15	192.168.1.254	TCP	60	65085 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
72	1961.6457391...	192.168.1.254	10.0.2.15	TCP	62	113 → 65085 [RST, ACK] Seq=1 Ack=1 Win=32768 Len=0
73	1961.6457393...	192.168.1.254	10.0.2.15	TCP	62	139 → 65085 [RST, ACK] Seq=1 Ack=1 Win=32768 Len=0
74	1961.6462931...	10.0.2.15	192.168.1.255	TCP	60	65085 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
75	1961.6468219...	10.0.2.15	192.168.1.254	TCP	60	65085 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
76	1961.6470641...	10.0.2.15	192.168.1.255	TCP	60	65085 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
77	1961.6483273...	10.0.2.15	192.168.1.254	TCP	60	65085 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
78	1961.6499176...	192.168.1.254	10.0.2.15	TCP	62	5900 → 65085 [RST, ACK] Seq=1 Ack=1 Win=32768 Len=0
79	1961.6502449...	10.0.2.15	192.168.1.255	TCP	60	65085 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
80	1961.6522595...	192.168.1.254	10.0.2.15	TCP	62	256 → 65085 [RST, ACK] Seq=1 Ack=1 Win=32768 Len=0
81	1961.6524279...	10.0.2.15	192.168.1.254	TCP	60	65085 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
82	1961.6547874...	10.0.2.15	192.168.1.255	TCP	60	65085 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
83	1961.6550159...	10.0.2.15	192.168.1.254	TCP	60	65085 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Observations:

S/N	Wireshark Network Capture Details	Explanation of observation
1	10.0.2.15 → 192.168.1.255 ICMP 44 Echo (ping) request 10.0.2.15 → 192.168.1.254 ICMP 44 Echo (ping) request - Source → 10.0.2.15 - Destination → 192.168.1.255 - Protocol → ICMP - Type → Ping request	Internet Control Message Protocol with a ping request to the default gateway IPs in the range being scanned. There was no response for the .255 while a reply was received from .254 within 57s.
2	10.0.2.15 → 192.168.1.255 TCP 60 64829 → 443 10.0.2.15 → 192.168.1.254 TCP 60 64829 → 443 - Source → 10.0.2.15 - Destination → 192.168.1.255 - Protocol → TCP - Type → from source port 64829 to destination port 443 (SYN)	A synchronization request (SYN) was sent from the Kali Linux server to port 443 on each of the 2 IPs addresses in the range being scanned; .254 & .255

S/N	Wireshark Network Capture Details	Explanation of observation
3	10.0.2.15 → 192.168.1.255 TCP 56 64829 → 80 [ACK]	
	10.0.2.15 → 192.168.1.254 TCP 56 64829 → 80 [ACK]	
	- Source → 10.0.2.15 - Destination → 192.168.1.255 - Protocol → <u>TCP</u> - Type → from source port 64829 to destination port 80 (ACK)	I observed that the acknowledgement (ACK) returned from port 80 of the destination. This may suggest a potential redirection or change in the port during the communication due to a few possibilities: - The server at the destination may be configured to redirect traffic from port 443 to port 80. Perhaps for load balancing or service availability. - There might be a network device, like a firewall or proxy, that is intercepting or modifying the traffic. - The server might be configured to handle traffic on port 443 but respond on port 80
S/N	Wireshark Network Capture Details	Explanation of observation
4	192.168.1.254 → 10.0.2.15 TCP 62 80 → 64829 [RST, ACK]	
	- Source → 192.168.1.254 - Destination → 10.0.2.15 - Protocol → <u>TCP</u> - Type → from source port 80 to destination port 64829 (RST, ACK)	This indicates a TCP packet sent from 192.168.1.254 to 10.0.2.15 with a reset (RST) and acknowledgment (ACK) flags set. This indicates that a previous connection, on port 80, is being reset and acknowledged.
S/N	Wireshark Network Capture Details	Explanation of observation
5	192.168.1.254 → 10.0.2.15 TCP 62 443 → 64829 [SYN, ACK]	
	- Source → 192.168.1.254 - Destination → 10.0.2.15 - Protocol → <u>TCP</u> - Type → from source port 443 to destination port 64829 (SYN, ACK)	This represents a TCP packet sent from 192.168.1.254 to 10.0.2.15 with the SYN (synchronize) and ACK (acknowledge) flags set. This typically means that the server at 192.168.1.254 is acknowledging the SYN request from the client at 10.0.2.15 and is initiating the establishment of a TCP connection.